Text Classification in NLP

CMPE346 · Group 2

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Overview

Group task

Text Classification

Metric

Accuracy & Weighted F1

Student	**Task**
Atakan Gül	Media Bias Detection
[Student 2]	[Topic]
[Student 3]	[Topic]

Models

BiLSTM

Bidirectional LSTM trained from scratch. Captures sequential context in both directions.

BERT `bert-base-uncased`

Transformer encoder pretrained via masked language modeling. Fine-tuned end-to-end.

RoBERTa `roberta-base`

Optimized BERT: removes next sentence prediction, uses dynamic masking and more data.

Task

Classify a news sentence as *biased* or *not biased*.

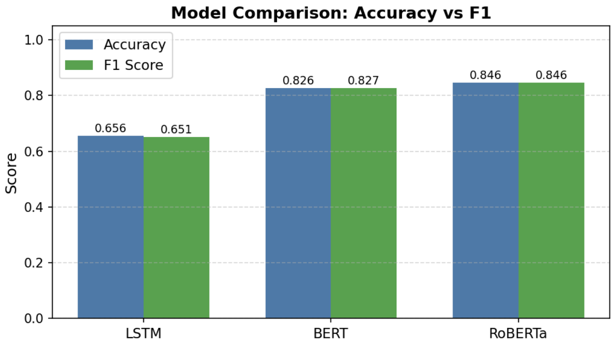
Dataset: BABE

- 4,121 expert-annotated sentences
- Diverse news outlets
- 3,121 train / 1,000 test

Atakan — Results

Model	Acc.	F1
BiLSTM	0.656	0.651
BERT	0.826	0.827
RoBERTa	**0.846**	**0.846**

- RoBERTa best: +19% over BiLSTM
- BERT close behind
- BiLSTM struggles on subtle bias



Atakan — Prediction Examples

Sentence	True	LSTM	BERT	RoBERTa
"Nearly 78% of Americans report concern about the climate..."	NB	NB ✓	NB ✓	NB ✓
"Companies like Nike and Google made Juneteenth a paid holiday."	NB	B ×	NB ✓	NB ✓
"None of Biden's Democrat counterparts challenged his claim of economic selflessness."	B	B ✓	NB ×	B ✓

NB = Not Biased   B = Biased   ✓ = correct   × = incorrect

Atakan — Conclusion

- Media bias is expressed through framing and tone — not facts
- Pretraining is critical: transformers capture patterns BiLSTM cannot
- RoBERTa's optimized pretraining gives it a clear edge over BERT

[Student 2] — [Topic]

Task

[Description]

Dataset: [Name]

Stats

Train / test split

[Student 2] — Results

Model	Acc.	F1
BiLSTM	—	—
BERT	—	—
RoBERTa	—	—

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[Figure]

[Student 2] — Conclusion

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[Student 3] — [Topic]

Task

[Description]

Dataset: [Name]

Stats

Train / test split

[Student 3] — Results

Model	**Acc.**	**F1**
BiLSTM	—	—
BERT	—	—
RoBERTa	—	—

[Figure]

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[Student 3] — Conclusion

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All Results

Task	Model	Acc.	F1
Media Bias	BiLSTM	0.656	0.651
	BERT	0.826	0.827
	RoBERTa	**0.846**	**0.846**
[Student 2]	BiLSTM	—	—
	BERT	—	—
	RoBERTa	—	—
[Student 3]	BiLSTM	—	—
	BERT	—	—
	RoBERTa	—	—

Conclusion

- RoBERTa consistently outperforms across all three tasks
- BERT and RoBERTa adapt well regardless of classification domain
- Pretraining is decisive — BiLSTM cannot compete on small datasets